

Timbre Assignment

Codebase KDoc & Architecture API Reference

Project Domain:	Media Editing & Playback (Audio / Video Trimming)
Architecture:	Clean Architecture + MVVM + Jetpack Compose + Hilt DI
Total Kotlin Source Files:	56 files documented
Document Scope:	Complete KDoc annotations across Data, Domain, Core, DI, and Presentation layers
Generation Date:	September 2026

Architecture Layer Breakdown

Layer Module	File Count	Description
Timbreassignment	56 files	Application logic layer

Complete Class & Function KDoc Reference

■ MainActivity.kt

Path: app/src/main/java/com/nutrino/timbreassignment/MainActivity.kt | **Package:** com.nutrino.timbreassignment

Declaration / Symbol	KDoc Documentation Details
@AndroidEntryPoint class MainActivity : ComponentActivity()	<i>Main activity entry point for the Timbre Assignment application. Configures the Android splash screen, enables edge-to-edge layout rendering, and attaches the root Jetpack Compose hierarchy with [TimbreAssignmentTheme] and [AppNavHost]. Annotated with [AndroidEntryPoint] for Hilt dependency injection initialization.</i>

■ ResultState.kt

Path: app/src/main/java/com/nutrino/timbreassignment/core/ResultState.kt | **Package:** com.nutrino.timbreassignment.core

Declaration / Symbol	KDoc Documentation Details
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■ MediaPlayerManager.kt

Path: app/src/main/java/com/nutrino/timbreassignment/core/media/MediaPlayerManager.kt | **Package:** com.nutrino.timbreassignment.core.media

Declaration / Symbol	KDoc Documentation Details
class MediaPlayerManager @Inject constructor()	Centralized manager wrapping Media3 [ExoPlayer] playback operations. Provides lifecycle methods for initializing media sources, controlling playback (play, pause, seek), and releasing player instances to prevent memory leaks. @property exoPlayer Injected [ExoPlayer] instance managed by Hilt DI.
fun getPlayer(): ExoPlayer	Retrieves the underlying [ExoPlayer] instance for Compose UI surface binding. @return The active [ExoPlayer] instance.
fun initializePlayer(uri: Uri)	Prepares the ExoPlayer with a media file identified by [uri]. Clears previous media items, prepares playback, and sets initial playback state. @param uri [Uri] pointing to the target audio or video resource.
fun play()	Begins or resumes media playback.
fun pause()	Pauses media playback.
fun seekTo(positionMs: Long)	Seeks playback to the specified time in milliseconds. @param positionMs Target playback timestamp in milliseconds.
fun releasePlayer()	Stops playback and releases ExoPlayer resources.

■ ResultState.kt

Path: app/src/main/java/com/nutrino/timbreassignment/core/states/ResultState.kt | **Package:** com.nutrino.timbreassignment.core.states

Declaration / Symbol	KDoc Documentation Details
sealed class ResultState	Encapsulates the execution state of an asynchronous operation or data query. @param T The type of data encapsulated upon success.
object Loading : ResultState()	Represents an ongoing or pending asynchronous operation.
data class Success(val data: T) : ResultState()	Represents a successful operation holding the resulting [data]. @property data The successfully retrieved data payload.
data class Error(val message: String) : ResultState()	Represents a failed operation holding an error [message]. @property message Descriptive failure explanation or localized error text.

■ PermissionUtils.kt

Path: app/src/main/java/com/nutrino/timbreassignment/core/utls/PermissionUtils.kt | **Package:** com.nutrino.timbreassignment.core.utls

Declaration / Symbol	KDoc Documentation Details
object PermissionUtils	<i>Helper utility for determining and checking runtime Android permissions required for media accessing and editing.</i>
fun getRequiredMediaPermissions(): Array	<i>Returns the array of runtime media permissions required based on the device Android API level. On Android 13+ (API level 33, Tiramisu), granular permissions (`READ_MEDIA_AUDIO`, `READ_MEDIA_VIDEO`, `READ_MEDIA_IMAGES`) are requested. On older versions, `READ_EXTERNAL_STORAGE` and `WRITE_EXTERNAL_STORAGE` are returned. @return Array of permission manifest strings.</i>
fun hasAllMediaPermissions(context: Context): Boolean	<i>Checks if all required media storage permissions have been granted by the user. @param context [Context] used to check runtime permission statuses. @return `true` if all permissions are granted, `false` otherwise.</i>

■ Song.kt

Path: app/src/main/java/com/nutrino/timbreassignment/data/dataclass/Song.kt | **Package:** com.nutrino.timbreassignment.data.dataclass

Declaration / Symbol	KDoc Documentation Details
data class Song (	<i>Data domain model representing an audio song file retrieved from MediaStore. @property id Unique MediaStore content ID. @property path File system path or URI string pointing to the audio file. @property size File size formatted as a string or byte count. @property album Album title associated with the song. @property title Song display title. @property artist Artist name. @property duration Track duration in milliseconds as a string. @property year Release year of the audio track. @property composer Track composer name. @property albumId MediaStore album identifier used to query album artwork. @property albumArt Optional preloaded album cover [Bitmap].</i>

■ Video.kt

Path: app/src/main/java/com/nutrino/timbreassignment/data/dataclass/Video.kt | **Package:** com.nutrino.timbreassignment.data.dataclass

Declaration / Symbol	KDoc Documentation Details
data class Video (	<i>Data domain model representing a video file retrieved from MediaStore. @property id Unique MediaStore video record ID. @property path Absolute file system path or content URI string. @property duration Total video duration in milliseconds formatted as a string. @property thumbnail Content URI or file path pointing to the video thumbnail image. @property fileName Display file name including extension. @property title Video title extracted from metadata or file name. @property folderName Containing directory name.</i>

■ MediaEditingRepoImpl.kt

Path: app/src/main/java/com/nutrino/timbreassignment/data/repoimpl/MediaEditingRepoImpl.kt | **Package:** com.nutrino.timbreassignment.data.repoimpl

Declaration / Symbol	KDoc Documentation Details
class MediaEditingRepoImpl @Inject constructor (	<i>Production implementation of [MediaEditingRepository]. Interacts with system [MediaStore] content resolvers to query device audio and video files, and executes FFmpeg commands via FFmpegKit to trim audio and video files efficiently without re-encoding. @property context Application context injected via Hilt for content resolver access.</i>

■ BaseApplication.kt

Path: app/src/main/java/com/nutrino/timbreassignment/di/baseclass/BaseApplication.kt | **Package:** com.nutrino.timbreassignment.di.baseclass

Declaration / Symbol	KDoc Documentation Details
@HiltAndroidApp class BaseApplication : Application()	<i>Base [Application] class for the Timbre Assignment app. Annotated with [@HiltAndroidApp] to trigger Hilt's code generation and set up the application-level dependency injection container.</i>

■ DiModule.kt

Path: app/src/main/java/com/nutrino/timbreassignment/di/modules/DiModule.kt | **Package:** com.nutrino.timbreassignment.di.modules

Declaration / Symbol	KDoc Documentation Details
@Singleton fun provideExoPlayer(@Applicatio nContext context: Context): ExoPlayer	Main Hilt dependency injection module installed in [SingletonComponent]. Provides application-level singletons including [ExoPlayer], [MediaPlayerManager], [MediaEditingRepository], and domain use cases. @Module @InstallIn(SingletonComponent::class) object DiModule { Provides a singleton instance of [ExoPlayer]. @param context Application context injected via Hilt. @return Configured [ExoPlayer] instance.
@Singleton fun provideMediaPlayerManager(ex oPlayer: ExoPlayer): MediaPlayerManager	Provides a singleton instance of [MediaPlayerManager]. @param exoPlayer Singleton [ExoPlayer] instance. @return [MediaPlayerManager] instance wrapping the ExoPlayer.
@Singleton fun provideMediaEditingRepository(	Provides [MediaEditingRepository] implementation. @param repositoryImpl Concrete repository implementation [MediaEditingRepoImpl]. @return [MediaEditingRepository] interface instance.
@Provides fun provideGetAllSongsUseCase(	Provides [GetAllSongsUseCase] instance. @param repository Injected [MediaEditingRepository]. @return [GetAllSongsUseCase] instance.
@Provides fun provideGetAllVideosUseCase(	Provides [GetAllVideosUseCase] instance. @param repository Injected [MediaEditingRepository]. @return [GetAllVideosUseCase] instance.
@Provides fun provideTrimAudioUseCase(	Provides [TrimAudioUseCase] instance. @param repository Injected [MediaEditingRepository]. @return [TrimAudioUseCase] instance.
@Provides fun provideTrimVideoUseCase(	Provides [TrimVideoUseCase] instance. @param repository Injected [MediaEditingRepository]. @return [TrimVideoUseCase] instance.

■ DispatcherModule.kt

Path: app/src/main/java/com/nutrino/timbreassignment/di/modules/DispatcherModule.kt | **Package:** com.nutrino.timbreassignment.di.modules

Declaration / Symbol	KDoc Documentation Details
@Provides fun provideIoDispatcher(): CoroutineDispatcher	Hilt module that provides coroutine dispatchers across the application. @Module @InstallIn(SingletonComponent::class) object DispatcherModule { Provides the IO [CoroutineDispatcher] for disk and network I/O operations. @return [Dispatchers.IO] dispatcher instance.

■ IoDispatcher.kt

Path: app/src/main/java/com/nutrino/timbreassignment/di/qualifiers/IoDispatcher.kt | **Package:** com.nutrino.timbreassignment.di.qualifiers

Declaration / Symbol	KDoc Documentation Details
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■ MediaEditingRepository.kt

Path: app/src/main/java/com/nutrino/timbreassignment/domain/repository/MediaEditingRepository.kt | **Package:** com.nutrino.timbreassignment.domain.repository

Declaration / Symbol	KDoc Documentation Details
interface MediaEditingRepository	<i>Domain repository contract for media querying and editing operations.</i>

■ GetAllSongsUseCase.kt

Path: app/src/main/java/com/nutrino/timbreassignment/domain/usecase/GetAllSongsUseCase.kt | **Package:** com.nutrino.timbreassignment.domain.usecase

Declaration / Symbol	KDoc Documentation Details
class GetAllSongsUseCase @Inject constructor(	<i>Domain use case for retrieving all audio songs from device storage. @property repository Injected [MediaEditingRepository].</i>

■ GetAllVideosUseCase.kt

Path: app/src/main/java/com/nutrino/timbreassignment/domain/usecase/GetAllVideosUseCase.kt | **Package:** com.nutrino.timbreassignment.domain.usecase

Declaration / Symbol	KDoc Documentation Details
class GetAllVideosUseCase @Inject constructor(	<i>Domain use case for retrieving all video files from device storage. @property repository Injected [MediaEditingRepository].</i>

■ TrimAudioUseCase.kt

Path: app/src/main/java/com/nutrino/timbreassignment/domain/usecase/TrimAudioUseCase.kt | **Package:** com.nutrino.timbreassignment.domain.usecase

Declaration / Symbol	KDoc Documentation Details
class TrimAudioUseCase @Inject constructor(	<i>Domain use case for executing audio trimming operations between start and end timestamps. @property repository Injected [MediaEditingRepository].</i>

■ TrimVideoUseCase.kt

Path: app/src/main/java/com/nutrino/timbreassignment/domain/usecase/TrimVideoUseCase.kt | **Package:** com.nutrino.timbreassignment.domain.usecase

Declaration / Symbol	KDoc Documentation Details
class TrimVideoUseCase @Inject constructor(	<i>Domain use case for executing video trimming operations between start and end timestamps. @property repository Injected [MediaEditingRepository].</i>

■ ErrorScreen.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/common/ErrorScreen.kt | **Package:** com.nutrino.timbreassignment.presentation.common

Declaration / Symbol	KDoc Documentation Details
@Composable fun ErrorScreen(	<i>Reusable full-screen error display showing an error icon, error message, and optional retry action button. @param message Localized or detailed error message string to display. @param onRetry Optional callback function invoked when the user clicks the retry button. @param modifier Optional [Modifier] for screen container layout customization.</i>

■ LoadingScreen.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/common/LoadingScreen.kt | **Package:** com.nutrino.timbreassignment.presentation.common

Declaration / Symbol	KDoc Documentation Details
@Composable fun LoadingScreen(	<i>Reusable full-screen indeterminate progress loading spinner view. @param modifier Optional [Modifier] for screen container styling.</i>

■ PermissionNotGrantedScreen.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/common/PermissionNotGrantedScreen.kt | **Package:** com.nutrino.timbreassignment.presentation.common

Declaration / Symbol	KDoc Documentation Details
@Composable fun PermissionNotGrantedScreen(	<i>Screen presented when required media permissions have not been granted by the user. Provides a clear explanation and a button trigger to request permission. @param onRequestPermissionClick Callback function triggered when the grant permission button is clicked. @param modifier Optional [Modifier] for screen container styling. @param title Header title explaining permission requirement. @param description Detailed permission usage explanation.</i>

■ AppNavHost.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/navigation/navhost/AppNavHost.kt |

Package: com.nutrino.timbreassignment.presentation.navigation.navhost

Declaration / Symbol	KDoc Documentation Details
@Composable fun AppNavHost(	<i>Top-level Jetpack Compose Navigation Host setting up type-safe navigation routes. Configures transitions between [HomeScreen], [GetAllAudioScreen], [AudioTrimmerScreen], [GetAllVideoScreen], and [VideoTrimmerScreen]. @param modifier Optional [Modifier] for container layout styling. @param navController [NavController] managing app backstack and navigation state.</i>

■ Routes.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/navigation/navroutes/Routes.kt |

Package: com.nutrino.timbreassignment.presentation.navigation.navroutes

Declaration / Symbol	KDoc Documentation Details
@Serializable object HOMESCREEN	<i>Navigation route destination for the Home Screen.</i>
@Serializable object ALLAUDIOSCREEN	<i>Navigation route destination for the Audio List Selection Screen.</i>
@Serializable object ALLVIDEOSCREEN	<i>Navigation route destination for the Video List Selection Screen.</i>
@Serializable data class AUDIOTRIMMERSCREEN(	<i>Navigation route destination for the Audio Trimmer Screen with song arguments. @property songPath Source song URI or file path. @property songTitle Display title of the audio track. @property songDurationMs Total track duration in milliseconds.</i>
@Serializable data class VIDEOTRIMMERSCREEN(	<i>Navigation route destination for the Video Trimmer Screen with video arguments. @property videoPath Source video URI or file path. @property videoTitle Display title of the video file. @property videoDurationMs Total video duration in milliseconds.</i>

■ AudioTrimmerContent.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/audiotrimmer/AudioTrimmerContent.kt |

Package: com.nutrino.timbreassignment.presentation.screens.audiotrimmer

Declaration / Symbol	KDoc Documentation Details
fun AudioTrimmerContent(	<i>No KDoc comment attached.</i>
fun AudioTrimmerContentIdlePreview()	<i>No KDoc comment attached.</i>

■ AudioTrimmerScreen.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/audiotrimmer/AudioTrimmerScreen.kt |

Package: com.nutrino.timbreassignment.presentation.screens.audiotrimmer

Declaration / Symbol	KDoc Documentation Details
@Composable fun AudioTrimmerScreen(	Screen component for trimming audio songs. Integrates ExoPlayer audio preview, range slider start/end timestamp selection, file naming, and Storage Access Framework (SAF) document export launchers. @param songPath Source audio song file path or URI string. @param songTitle Title of the song to trim. @param songDurationMs Total audio track duration in milliseconds. @param onBackPressed Pop backstack navigation callback. @param mediaViewModel [MediaViewModel] managing ExoPlayer audio playback. @param viewModel [MediaEditingViewModel] managing FFmpeg audio trim requests.

■ AudioPlayerPreview.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/audiotrimmer/components/AudioPlayerPreview.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.audiotrimmer.components

Declaration / Symbol	KDoc Documentation Details
fun AudioPlayerPreview(	No KDoc comment attached.
fun AudioPlayerPreviewPreview(	No KDoc comment attached.

■ OutputFileNameInput.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/audiotrimmer/components/OutputFileNameInput.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.audiotrimmer.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun OutputFileNameInput(	Text field component for specifying the trimmed output file name. @param fileName Current file name string value. @param onFileNameChange Text modification callback. @param modifier Layout modifier.

■ TrimRangeSlider.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/audiotrimmer/components/TrimRangeSlider.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.audiotrimmer.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun TrimRangeSlider(	<i>Interactive range slider component for picking start and end trim timestamps. @param startMs Selected start time offset in milliseconds. @param endMs Selected end time offset in milliseconds. @param totalDurationMs Total media duration in milliseconds. @param isPlaying Playback state boolean. @param onPlayPauseClick Toggle playback callback. @param onRangeChange Range change callback emitting updated (startMs, endMs). @param modifier Layout modifier.</i>

■ TrimStateViews.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/audiotrimmer/components/TrimStateViews.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.audiotrimmer.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun TrimLoadingView(	<i>Loading indicator card view displayed while trimming is in progress. @param modifier Layout modifier.</i>
@Composable fun TrimSuccessView(	<i>Success card view displayed when trimming completes successfully. @param outputPath Destination URI path string. @param onResetClick Reset trim flow callback. @param onSaveToFolderClick SAF export action callback. @param modifier Layout modifier.</i>
@Composable fun TrimErrorView(	<i>Error card view displayed when media trimming fails. @param message Error explanation message. @param onRetryClick Retry callback. @param modifier Layout modifier.</i>

■ TrimAudioState.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/audiotrimmer/states/TrimAudioState.kt |

Package: com.nutrino.timbreassignment.presentation.screens.audiotrimmer.states

Declaration / Symbol	KDoc Documentation Details
sealed interface TrimAudioState	<i>UI State interface for the audio trimming operation.</i>
object Idle : TrimAudioState	<i>Idle state before trimming is triggered.</i>
object Loading : TrimAudioState	<i>Loading state while FFmpeg executes audio trimming.</i>
data class Success(val outputPath: String) : TrimAudioState	<i>Success state holding the trimmed output file destination URI/path. @property outputPath Destination URI string.</i>
data class Error(val message: String) : TrimAudioState	<i>Error state holding trim failure message. @property message Failure details.</i>

■ GetAllAudioContent.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallaudio/GetAllAudioContent.kt

| **Package:** com.nutrino.timbreassignment.presentation.screens.getallaudio

Declaration / Symbol	KDoc Documentation Details
fun GetAllAudioContent(	No KDoc comment attached.
var searchQuery by rememberSaveable	No KDoc comment attached.
val filteredSongs	No KDoc comment attached.
val query	No KDoc comment attached.
fun GetAllAudioContent_LoadingPreview()	No KDoc comment attached.
fun GetAllAudioContent_SuccessPreview()	No KDoc comment attached.
fun GetAllAudioContent_ErrorPreview()	No KDoc comment attached.

■ GetAllAudioScreen.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallaudio/GetAllAudioScreen.kt

| **Package:** com.nutrino.timbreassignment.presentation.screens.getallaudio

Declaration / Symbol	KDoc Documentation Details
fun GetAllAudioScreen(	No KDoc comment attached.
val context	No KDoc comment attached.
var hasPermission by remember	No KDoc comment attached.
val permissionLauncher	No KDoc comment attached.
val granted	No KDoc comment attached.
val state by viewModel.getAllSongsState.collectAsStateWithLifecycle()	No KDoc comment attached.
fun GetAllAudioScreenPreview()	No KDoc comment attached.

■ NoAudioFoundView.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallaudio/components/NoAudio

FoundView.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.getallaudio.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun NoAudioFoundView(	Placeholder view displayed when no audio files are found or match search queries. @param modifier Layout modifier. @param message Headline message text. @param description Subtitle descriptive explanation. @param onRefresh Optional refresh action callback.

■ SongItem.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallaudio/components/SongItem.kt |

Package: com.nutrino.timbreassignment.presentation.screens.getallaudio.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun SongItem(	List item card displaying song details including title and artist. @param song The [Song] data object. @param onClick Click callback triggered on item selection. @param modifier Layout modifier.

■ GetAllSongState.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallaudio/states/GetAllSongState.kt |

Package: com.nutrino.timbreassignment.presentation.screens.getallaudio.states

Declaration / Symbol	KDoc Documentation Details
sealed interface GetAllSongState	UI State interface for the Get All Songs list query workflow.
object Idle : GetAllSongState	Idle state before query execution.
object Loading : GetAllSongState	Loading state while songs are being queried from MediaStore.
data class Success(val data: List) : GetAllSongState	Success state holding the queried list of songs. @property data List of retrieved [Song] instances.
data class Error(val message: String) : GetAllSongState	Error state holding failure explanation. @property message Error details.

■ GetAllVideoContent.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallvideo/GetAllVideoContent.kt

| Package: com.nutrino.timbreassignment.presentation.screens.getallvideo

Declaration / Symbol	KDoc Documentation Details
fun GetAllVideoContent(	No KDoc comment attached.
var searchQuery by rememberSaveable	No KDoc comment attached.
val filteredVideos	No KDoc comment attached.
val query	No KDoc comment attached.
fun GetAllVideoContent_LoadingPreview()	No KDoc comment attached.
fun GetAllVideoContent_SuccessPreview()	No KDoc comment attached.
fun GetAllVideoContent_ErrorPreview()	No KDoc comment attached.

■ GetAllVideoScreen.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallvideo/GetAllVideoScreen.kt |

Package: com.nutrino.timbreassignment.presentation.screens.getallvideo

Declaration / Symbol	KDoc Documentation Details
fun GetAllVideoScreen(	No KDoc comment attached.
val context	No KDoc comment attached.
var hasPermission by remember	No KDoc comment attached.
val permissionLauncher	No KDoc comment attached.
val granted	No KDoc comment attached.
val state by viewModel.getAllVideosState.collectAsStateWithLifecycle()	No KDoc comment attached.
fun GetAllVideoScreenPreview()	No KDoc comment attached.

■ NoVideoFoundView.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallvideo/components/NoVideoFoundView.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.getallvideo.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun NoVideoFoundView(	Empty placeholder view displayed when no videos are found or match search criteria. @param modifier Layout modifier. @param message Header message. @param description Subtitle description string. @param onRefresh Refresh callback.

■ VideoItem.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallvideo/components/VideoItem.kt |

Package: com.nutrino.timbreassignment.presentation.screens.getallvideo.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun VideoItem(	List item card displaying video details including title and containing folder. @param video The [Video] model object. @param onClick Selection click callback. @param modifier Layout modifier.

■ GetAllVideoState.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/getallvideo/states/GetAllVideoState.kt |

Package: com.nutrino.timbreassignment.presentation.screens.getallvideo.states

Declaration / Symbol	KDoc Documentation Details
sealed interface GetAllVideoState	<i>UI State interface for the Get All Videos list query workflow.</i>
object Idle : GetAllVideoState	<i>Idle state before query execution.</i>
object Loading : GetAllVideoState	<i>Loading state while videos are being queried from MediaStore.</i>
data class Success(val data: List) : GetAllVideoState	<i>Success state holding the queried list of videos. @property data List of retrieved [Video] instances.</i>
data class Error(val message: String) : GetAllVideoState	<i>Error state holding failure explanation. @property message Error details.</i>

■ HomeScreen.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/homescreen/HomeScreen.kt |

Package: com.nutrino.timbreassignment.presentation.screens.homescreen

Declaration / Symbol	KDoc Documentation Details
@Composable fun HomeScreen(	<i>Root Composable for the Home Dashboard screen. Hosts top app bar header, bottom navigation bar switching between Audio, Video, and Settings tabs, and feature trigger cards for launching Audio and Video trimming tools. @param modifier Layout modifier. @param onTrimAudioClick Navigation trigger callback for audio trimming flow. @param onTrimVideoClick Navigation trigger callback for video trimming flow.</i>

■ FeatureCard.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/homescreen/components/FeatureCard.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.homescreen.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun FeatureCard(	<i>Interactive card UI component representing a primary feature action on the home screen. @param title Title text for the feature action (e.g., "Trim Audio"). @param icon Vector icon representing the action. @param onClick Callback triggered on card click. @param modifier Layout modifier.</i>

■ HomeScreenBottomBar.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/homescreen/components/HomeScreenBottomBar.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.homescreen.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun HomeScreenBottomBar(	<i>Bottom navigation bar allowing users to switch between home screen tabs. @param selectedTab The currently active [HomeTab]. @param onTabSelected Callback function invoked when a navigation tab is tapped.</i>

■ HomeScreenHeader.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/homescreen/components/HomeScreenHeader.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.homescreen.components

Declaration / Symbol	KDoc Documentation Details
fun HomeScreenHeader()	<i>No KDoc comment attached.</i>
fun HomeScreenHeaderPreview()	<i>No KDoc comment attached.</i>

■ HomeTab.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/homescreen/model/HomeTab.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.homescreen.model

Declaration / Symbol	KDoc Documentation Details
enum class HomeTab(	<i>Enumeration representing bottom navigation tabs on the Home screen. @property title Display title label. @property icon Vector icon associated with the tab.</i>

■ SettingsScreen.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/settings/SettingsScreen.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.settings

Declaration / Symbol	KDoc Documentation Details
@Composable fun SettingsScreen(	<i>Settings tab screen displaying app permission statuses and shortcuts to open System Application Details Settings. @param modifier Layout modifier.</i>
@Composable fun PermissionStatusItem(	<i>List item card displaying individual permission status and navigation affordance. @param title Permission description label. @param icon Icon representing the permission category. @param isAllowed Boolean status of the permission grant. @param onClick Click handler to launch System Settings. @param modifier Layout modifier.</i>

■ VideoTrimmerContent.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/videotrimmer/VideoTrimmerContent.kt |

Package: com.nutrino.timbreassignment.presentation.screens.videotrimmer

Declaration / Symbol	KDoc Documentation Details
fun VideoTrimmerContent(	No KDoc comment attached.
fun VideoTrimmerContentIdlePreview()	No KDoc comment attached.

■ VideoTrimmerScreen.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/videotrimmer/VideoTrimmerScreen.kt |

Package: com.nutrino.timbreassignment.presentation.screens.videotrimmer

Declaration / Symbol	KDoc Documentation Details
@Composable fun VideoTrimmerScreen(	Screen component for trimming video files. Provides ExoPlayer video player preview, range slider start/end time selection, output file naming, and Storage Access Framework document export. @param videoPath Source video file path or URI string. @param videoTitle Display title of the video file. @param videoDurationMs Total video duration in milliseconds. @param onBackClick Navigation pop backstack action. @param mediaViewModel [MediaViewModel] managing video preview playback. @param viewModel [MediaEditingViewModel] executing video trimming requests.

■ VideoOutputFileNameInput.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/videotrimmer/components/VideoOutputFileNameInput.kt | **Package:**

com.nutrino.timbreassignment.presentation.screens.videotrimmer.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun VideoOutputFileNameInput(	Text field component for entering the output video file name. @param fileName Current file name string. @param onFileNameChange Text input modification callback. @param modifier Layout modifier.

■ VideoPlayerPreview.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/videotrimmer/components/VideoPlayerPreview.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.videotrimmer.components

Declaration / Symbol	KDoc Documentation Details
fun VideoPlayerPreview(	No KDoc comment attached.
fun VideoPlayerPreviewPreview()	No KDoc comment attached.

■ VideoTrimStateViews.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/screens/videotrimmer/components/VideoTrimStateViews.kt | **Package:** com.nutrino.timbreassignment.presentation.screens.videotrimmer.components

Declaration / Symbol	KDoc Documentation Details
@Composable fun VideoTrimLoadingView(	Loading indicator view for ongoing video trimming operations. @param modifier Layout modifier.
@Composable fun VideoTrimSuccessView(	Success view card displayed when video trimming completes. @param outputPath Destination output URI path string. @param onResetClick Reset trim action callback. @param onSaveToFolderClick SAF document export callback. @param modifier Layout modifier.
@Composable fun VideoTrimErrorView(	Error view card displayed when video trimming fails. @param message Error description message. @param onRetryClick Retry action callback. @param modifier Layout modifier.

■ TrimVideoState.kt

Path:

app/src/main/java/com/nutrino/timbreassignment/presentation/screens/videotrimmer/states/TrimVideoState.kt |

Package: com.nutrino.timbreassignment.presentation.screens.videotrimmer.states

Declaration / Symbol	KDoc Documentation Details
sealed interface TrimVideoState	UI State interface for the video trimming operation.
object Idle : TrimVideoState	Idle state before video trimming.
object Loading : TrimVideoState	Loading state while FFmpeg executes video trimming.
data class Success(val outputPath: String) : TrimVideoState	Success state holding the trimmed output video destination path/URI. @property outputPath Destination URI string.
data class Error(val message: String) : TrimVideoState	Error state holding trim failure details. @property message Failure explanation.

■ MediaEditingViewModel.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/viewmodel/MediaEditingViewModel.kt |

Package: com.nutrino.timbreassignment.presentation.viewmodel

Declaration / Symbol	KDoc Documentation Details
@HiltViewModel class MediaEditingViewModel @Inject constructor()	Main [ViewModel] responsible for orchestrating media list queries and media trimming workflows. Exposes UI states via [StateFlow]s for song listings, video listings, audio trimming, and video trimming. @property getAllSongsUseCase Injected use case for querying device songs. @property getAllVideosUseCase Injected use case for querying device videos. @property trimAudioUseCase Injected use case for audio trimming. @property trimVideoUseCase Injected use case for video trimming. @property ioDispatcher Injected IO dispatcher for background execution.
val getAllSongsState: StateFlow	StateFlow emitting the current list state of song tracks.
val getAllVideosState: StateFlow	StateFlow emitting the current list state of video files.
val trimAudioState: StateFlow	StateFlow emitting the current audio trim execution status.
val trimVideoState: StateFlow	StateFlow emitting the current video trim execution status.
fun getAllSongs()	Fetches all audio songs from device storage.
fun getAllVideos()	Fetches all video files from device storage.
fun trimAudio()	Trims an audio file between [startMs] and [endMs]. @param inputPath Source file path. @param outputPath Target document destination URI. @param startMs Start time in milliseconds. @param endMs End time in milliseconds.
fun trimVideo()	Trims a video file between [startMs] and [endMs]. @param inputPath Source file path. @param outputPath Target document destination URI. @param startMs Start time in milliseconds. @param endMs End time in milliseconds.
fun resetTrimAudioState()	Resets the audio trim state to [TrimAudioState.Idle].
fun resetTrimVideoState()	Resets the video trim state to [TrimVideoState.Idle].

MediaViewModel.kt

Path: app/src/main/java/com/nutrino/timbreassignment/presentation/viewmodel/MediaViewModel.kt | **Package:** com.nutrino.timbreassignment.presentation.viewmodel

Declaration / Symbol	KDoc Documentation Details
@HiltViewModel class MediaViewModel @Inject constructor()	<i>[ViewModel] managing playback lifecycle and player controls for audio and video media previews. Interacts with [MediaPlayerManager] to expose playback commands (play, pause, seek, release). @property mediaEditingRepository Injected repository instance. @property mediaPlayerManager Injected player manager instance wrapping ExoPlayer.</i>
fun initializePlayer(uri: Uri)	<i>Initializes the player with the given media source [uri]. @param uri [Uri] pointing to the media file.</i>
fun getPlayer(): ExoPlayer	<i>Retrieves the active [ExoPlayer] instance. @return Underlying [ExoPlayer].</i>
fun play()	<i>Starts or resumes playback.</i>
fun pause()	<i>Pauses active playback.</i>
fun seekTo(positionMs: Long)	<i>Seeks playback to [positionMs]. @param positionMs Target timestamp in milliseconds.</i>
fun releasePlayer()	<i>Stops and releases player resources.</i>

Color.kt

Path: app/src/main/java/com/nutrino/timbreassignment/ui/theme/Color.kt | **Package:** com.nutrino.timbreassignment.ui.theme

Declaration / Symbol	KDoc Documentation Details
val AppBlack	<i>Solid black color token used for app background and dark surfaces.</i>
val AppWhite	<i>Solid white color token used for text and iconography on dark background.</i>
val CrimsonAccent	<i>Crimson red accent color token used for primary action items, highlights, and borders.</i>

■ Theme.kt

Path: app/src/main/java/com/nutrino/timbreassignment/ui/theme/Theme.kt | **Package:** com.nutrino.timbreassignment.ui.theme

Declaration / Symbol	KDoc Documentation Details
@Composable fun TimbreAssignmentTheme(	<i>Crimson and dark surface color palette for Material 3 styling across Timbre Assignment. private val CrimsonColorPalette = darkColorScheme(primary = CrimsonAccent, onPrimary = AppBlack, primaryContainer = AppBlack, onPrimaryContainer = AppWhite, secondary = CrimsonAccent, onSecondary = AppBlack, secondaryContainer = AppBlack, onSecondaryContainer = AppWhite, tertiary = CrimsonAccent, onTertiary = AppBlack, tertiaryContainer = AppBlack, onTertiaryContainer = AppWhite, background = AppBlack, onBackground = AppWhite, surface = AppBlack, onSurface = AppWhite, surfaceVariant = AppBlack, onSurfaceVariant = AppWhite, inverseSurface = AppBlack, inverseOnSurface = AppWhite, inversePrimary = CrimsonAccent, error = Color(0xFFFFF5252), onError = AppBlack, errorContainer = AppBlack, onErrorContainer = Color(0xFFFFF5252), outline = CrimsonAccent, outlineVariant = CrimsonAccent, scrim = AppBlack) Custom Material 3 theme wrapper applying crimson dark palette and typography to the application UI tree. @param content Composable UI hierarchy wrapped within this theme.</i>

■ Type.kt

Path: app/src/main/java/com/nutrino/timbreassignment/ui/theme/Type.kt | **Package:** com.nutrino.timbreassignment.ui.theme

Declaration / Symbol	KDoc Documentation Details
val Typography	<i>Material 3 typography definitions configured for the application.</i>